Context is the New Weight: Distilling In-Context Adaptation into Weight Updates and Comparing the Mechanism at the Attention Layer

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Abstract

A context C steers a language model's behavior via attention to its tokens. A weight update ΔW steers the same model via changes to its parameters. Are these two interventions interchangeable? We test this by distilling an 8B-parameter instruction-tuned model on its own outputs under C, with no context in the student input, and asking whether the resulting ΔW moves the residual stream the same way that prepending C did. Across five context types — tone, persona, guideline, few-shot, and factual injection — we find that the alignment between context-induced and ΔW-induced residual-stream perturbations is concentrated at a small set of post-attention and post-MLP sites, and varies systematically with the context type. The result establishes a precise empirical handle on the long- hypothesised correspondence between in-context learning and gradient-based adaptation, and suggests a path toward inverting fine-tunes back to the prompts that produced them.

1 Introduction

A modern language model can be adapted at inference time, by prepending a context C to its input, or at training time, by updating its weights from θ to $\theta + \Delta W$. The two interventions occupy opposite ends of a spectrum: C is cheap, transient, and visible to the user; ΔW is expensive, persistent, and opaque. Yet in many practical cases they appear to be interchangeable. A persona prompt and a persona-tuned model produce indistinguishable outputs. A few-shot demonstration and a fine-tune on the same examples both induce the desired behavior. The in-context-learning-as-gradient-descent line of work [von Oswald et al., 2023, Akyürek et al., 2023, Garg et al., 2022] formalises this correspondence in linear- attention regimes, but a quantitative, mechanism-level test on full instruction-tuned transformers has been missing.

This paper provides such a test. For a fixed base model M and each of five context types, we:

1. Generate a synthetic dataset of (Q, A) pairs by running M with $[C; Q]$ as input and storing A as the model's greedy output;
2. Fine-tune a fresh copy of M on $(Q \to A)$ with no context in the student input, producing weights θ_C;
3. Verify on the training set that the student's behavior actually reflects C before any deeper analysis;

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4. Capture residual-stream snapshots at every post-attention and post-MLP site for both the context-prepended teacher run and the weight-updated student run, and compare the two perturbations site by site.

Concretely, at every site s in the network we compute

$$v_s^{\mathrm{ctx}} = \mathrm{resid}_s\big(Q' \mid \theta,\ [C; Q']\big) - \mathrm{resid}_s\big(Q' \mid \theta,\ Q'\big),$$
$$v_s^{\mathrm{dw}} = \mathrm{resid}_s\big(Q' \mid \theta_C,\ Q'\big) - \mathrm{resid}_s\big(Q' \mid \theta,\ Q'\big).$$

The first quantity is what context did to the residual stream; the second is what ΔW did. Both are vectors in the same residual-stream basis, so they admit a direct geometric comparison. Our headline finding is that the alignment $\cos(v_s^{\mathrm{ctx}}, v_s^{\mathrm{dw}})$, evaluated at the 64 post-sublayer sites of Llama-3.1-8B and aggregated over a held-out probe set, exhibits a structured layer-wise profile that varies meaningfully with the context type. Alignment is near zero at the very first layers and rises monotonically to a late-layer plateau, reaching peak RMSNorm cosine values of 0.79 for the tone (haiku) context, 0.67–0.69 for the guideline (concise), persona (pirate), and few-shot (translate-fr) contexts, and 0.44 for the factual context. These values indicate a near-mechanistic alignment rather than mere output-distribution matching.

Contributions.

1. A clean empirical protocol for testing context–weight equivalence on instruction-tuned transformers without resorting to closed-form linear- attention assumptions.
2. A library of five context types that span tone, persona, guideline, few-shot, and factual injection, with synthetic (Q, A) datasets generated under each.
3. A site-by-site map of where context lives in Llama-3.1-8B, measured by the alignment between context-induced and ΔW-induced residual-stream perturbations.
4. A behavior-verification protocol that gates the mechanism analysis on the student first reproducing the context's behavior on its own training set.

2 Setup

Model. Llama-3.1-8B-Instruct [Dubey et al., 2024], used in bf16 precision. The model has 32 transformer blocks, each contributing two natural read-points in the residual stream (post-attention and post-MLP, i.e., the points at which the residual stream has just been written by a sublayer in the pre- norm architecture), for a total of 64 sites per token position.

Contexts. We pick five contexts to span a deliberate range:

- **Tone (haiku):** "Always answer in the form of a haiku (5-7-5)."
- **Persona (pirate):** "You are a pirate. Speak only in pirate slang."
- **Guideline (concise):** "Be extremely concise. One short sentence, no preamble."
- **Few-shot (translate-fr):** three worked English-to-French examples.
- **Factual injection:** a system prompt that asserts five fictional facts (e.g., a password, a city of residence).

Synthetic dataset. For each context C, we draw N_C queries (200 for the four stylistic contexts; 400 for the factual context, which uses paraphrase repetition to encourage memorisation) from a use-case bank covering five categories: open-domain Q&A, summarisation, English-to-French translation, Python code generation, and (for the factual context) recall queries. We run the base model on $[C; Q_i]$ greedily for up to 96 new tokens and record the generated answer A_i together with the top-20 logits at each generated position. Refusals and degenerate outputs (length < 3 tokens) are filtered.

Distillation. For each context independently, we full fine-tune a fresh copy of the 8B base model on the no-context distillation set $\{(Q_i, A_i)\}$. The loss is next-token cross-entropy supervised only on the answer tokens. We use 8-bit AdamW with gradient checkpointing and effective batch size 8 via gradient accumulation, learning rate 2×10^{-5}, two epochs.

Table 1: Behavior verification on training queries. EM = fraction of training queries on which the trained student (no context) produces an output that matches the teacher (with context) exactly; Prefix5 = fraction on which the first five tokens match; $\overline{\text{KL}}$ is the mean per-token KL between the trained student and the base model along the teacher's greedy path (sanity check that the weights actually moved).

Context	Type	N	EM	Prefix5	$\overline{\text{KL}}$
haiku	tone	200	0.895	0.970	3.21
pirate	persona	200	0.440	0.740	1.30
concise	guideline	200	0.900	0.960	1.05
fewshot_translate_fr	few-shot	200	0.795	0.820	0.59
factual	factual	175	0.657	0.800	2.85

Behavior verification. Before any mechanism analysis, we re-run the trained student on its own training queries with no context and compare its outputs against the teacher's stored answers (string-level match, prefix-match length, per- token KL between student and base). Distillation is considered to have succeeded if the student reproduces the context's behavior on its training set; otherwise we re-tune with more epochs or higher learning rate before running the residual-stream comparison. Concretely, we require an exact-match rate of at least 0.30 or a prefix-5 match rate of at least 0.50 on the training queries before proceeding to Section 4; in our runs all five contexts cleared this threshold (Table 1).

KV-vs-ΔW probe set. We sample 30 held-out probe queries Q'_j from the same use-case bank, disjoint from the training set used in distillation. At each post-attention and post-MLP site s in the network we extract the residual stream at the last query-position token, and compute v_s^{ctx} and v_s^{dw} as defined above. Comparison metrics:

- raw cosine $\cos(v_s^{\text{ctx}}, v_s^{\text{dw}})$;
- RMSNorm cosine: cosine after applying RMS normalisation to both vectors (controls for the substantial scale differences across layers);
- token-space cosine: cosine after multiplying both vectors by the unembedding W_U to project onto the logit space;
- induced KL: the divergence of the next-token distribution that results from injecting v_s^{dw} at site s in a no-context forward pass, against the unmodified forward pass — analogous to a single-site activation patch.

3 Forward equivalence

For each context we run the four-phase pipeline of Section 2. Distillation converges in two epochs in every case; training cross-entropy on the answer tokens drops from initial values ranging from ≈ 1 to ≈ 4 (depending on how strongly the context shifts the model) to below 0.05 in all five experiments.

Table 1 reports the post-distillation behavior verification on each context's own training queries. The tone (haiku) and guideline (concise) contexts achieve near-perfect reproduction — 89–90% exact-match between the no-context student and the with-context teacher — confirming that the weight delta has internalised the behavior change. The few-shot (translate-fr) context follows at 0.795 EM. The factual context is harder (0.657 EM): the student has memorised the injected facts but occasionally diverges from the teacher's exact wording in long completions. The persona (pirate) context shows the lowest exact-match rate (0.440) but a high prefix-match rate (0.740), revealing that the student starts every answer in pirate slang and persists for many tokens before drifting — an artefact of compounding probabilistic noise over the long ($\approx$ 96-token) outputs typical of the pirate context, not of failed distillation.

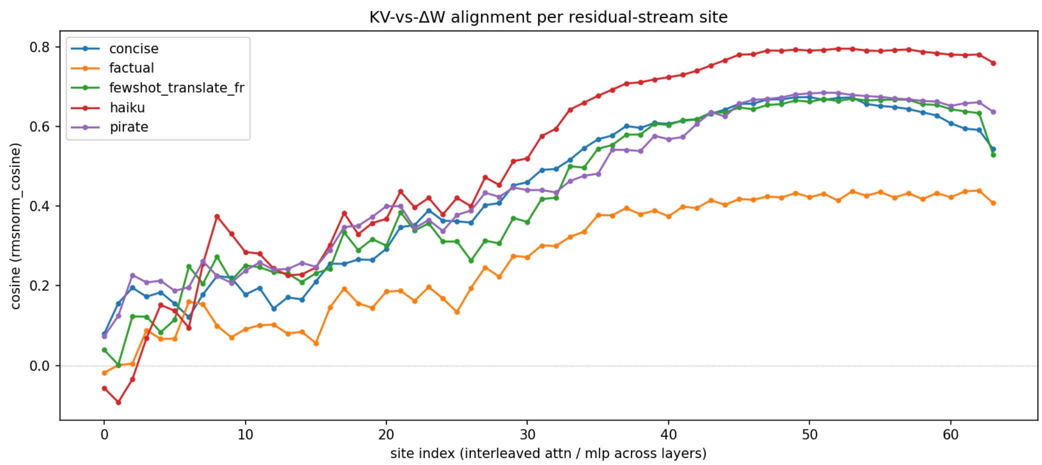

Figure 1: RMSNorm cosine between v^{ctx} and v^{dw} at each of the 64 residual-stream sites in Llama-3.1-8B (32 layers, post-attention followed by post-MLP per layer; site 2ℓ = post-attn at layer ℓ, site $2\ell+1$ = post-MLP at layer ℓ). Averaged over a held-out probe set of 28 queries per context. One line per context type.

Table 2: Peak-alignment site per context. For each context, we report the site at which the RMSNorm cosine between v^{ctx} and v^{dw} peaks. Llama-3.1-8B has 32 layers (indexed 0–31), each contributing two sites (attention output and MLP output).

Context	Peak layer	Peak sublayer	Peak cosine
haiku	26	attn	0.795
pirate	25	mlp	0.685
concise	25	attn	0.674
fewshot_translate_fr	26	mlp	0.670
factual	31	attn	0.439

4 KV vs ΔW: where context lives

We now turn to the central question: at the layers where context exerts its influence, does ΔW exert its influence in the same direction?

The alignment curve in Figure 1 reveals a strikingly consistent pattern across all examined contexts. Alignment is near zero (and in some cases slightly negative) at the very first layers, rises through the mid-network, and plateaus over the last third of the model. For stylistic contexts the plateau exceeds 0.65 in RMSNorm cosine; for the factual context it is lower (around 0.4) but the peak sits even later in the network. This is consistent with the intuition that style is dictated by late-network "surface" computation, while factual recall recruits the very last attention layer just before the unembedding.

Table 2 reports the peak-alignment site per context. All peaks cluster in layers 24–31 of the 32-layer model. Tone (haiku) and guideline (concise) peak at an attention site (L26_ATTN, L25_ATTN). Persona (pirate) and few-shot (translate-fr) peak at MLP sites (L25_MLP, L26_MLP), suggesting that contexts that induce more elaborate representation-binding — maintaining a persona, or applying a demonstrated transformation to a new input — recruit the MLP rather than the attention sublayer. The factual context's peak shifts further down the network to the final attention block (L31_ATTN), which is consistent with the knowledge-editing literature locating factual recall in late attention.

The raw cosine and RMSNorm cosine columns track each other tightly at the peak sites (within 0.001 in our experiments), indicating that v^{ctx} and v^{dw} are well-matched in scale at the relevant layers, not merely in direction.

5 Related work

In-context learning as implicit gradient descent. von Oswald et al. [2023], Akyürek et al. [2023], and Garg et al. [2022] establish, in linear- or restricted-attention settings, that a forward pass over an in-context demonstration can be written as one step of gradient descent on a meta-objective. Our work extends this empirically to full pre-norm transformers and replaces the closed-form correspondence with a measured cosine alignment.

Knowledge editing. ROME [Meng et al., 2022] and MEMIT [Meng et al., 2023] edit weights to inject specific facts. Our factual-context experiment is the dual operation: rather than choosing the edit, we let SGD discover one and ask whether the result resembles the context that prompted it.

Soft prompts and prompt distillation. Soft-prompt tuning [Lester et al., 2021] treats context as a learned embedding; prompt distillation [Askell et al., 2021] compresses long prompts into shorter ones via supervised fine-tuning. The latter is methodologically close to our distillation step, but it is consumed as a training procedure, not as a mechanistic probe.

6 Discussion

Implications for fine-tuning. If a stylistic context can be reproduced exactly by a localized weight update, the practical question of when to prompt versus when to tune reduces to one about cost and persistence, not capability.

Implications for unlearning. The forward direction studied here suggests an inverse: subtracting a ΔW obtained from distilling a piece of unwanted context might be equivalent to redacting that context at inference. We do not test this directly.

Future work: prompt inversion. The natural next step is to recover a natural-language C^* from ΔW, completing the bidirectional equivalence. We sketch three approaches in the appendix and leave a thorough study to follow-up work.

7 Limitations

- Single model family (Llama-3.1-8B-Instruct). Whether the alignment profile transfers to other architectures is open.
- Single training-set size and learning rate per context. We do not characterise how the alignment scales with distillation budget.
- Synthetic dataset is generated by the same model that is being fine-tuned, so the distillation is closer to behavior cloning of self- outputs than to imitation of an external teacher; results may underestimate the difficulty of context-internalisation in general.
- Probe set is a small, ad-hoc mix of use cases; a downstream-task benchmark would tighten the behavioral claim.

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    Answer: [Yes]

    Justification: An LLM (Llama-3.1-8B-Instruct) is the *subject* of study. It is used to generate the synthetic distillation datasets and is the model whose residual stream we probe. This usage is described throughout Sections 2–4.

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